

Chile-ED-Resp: A Curated and Reproducible Weekly Hospital Dataset for Respiratory Emergency-Demand Forecasting in Chile

Wilman Balcázar-Quimi ¹, Carla Angulo ^{2,*}, Cristian Cornejo-Gaete ³, Brandon Salinas-Neira ², Jorge Fernández-Bastías ², Miguel Tupac-Yupanqui ⁴, and Cristian Vidal-Silva ^{5,*}

- ¹ Facultad de Salud y Servicios Sociales, Universidad Estatal de Milagro, Milagro 092301, Guayas, Ecuador; wbalcazarq@unemi.edu.ec
- ² Escuela de Ciencias Digitales y Tecnologías Avanzadas, Universidad Bernardo O'Higgins, Santiago 8370846, Chile; carla.angulo@ubo.cl, {brandon.salinas, jorge.fernandez}@pregrado.ubo.cl
- ³ Facultad de Ingeniería, Ciencia y Tecnología, Universidad Bernardo O'Higgins, Santiago 8370846, Chile; cristian.cornejo@ubo.cl
- ⁴ EAP Ingeniería de Sistemas e Informática, Universidad Continental, Huancayo 12000, Peru; mtupac@continental.edu.pe
- ⁵ Facultad de Ingeniería y Negocios, Universidad de Las Américas, Manuel Montt 948, Providencia, Santiago 7500975, Chile; cristian.vidal.silva@edu.udla.cl
- * Correspondence: carla.angulo@ubo.cl (C.A.); cristian.vidal.silva@edu.udla.cl (C.V.-S.)

Citation: Balcázar-Quimi, W.; Angulo, C.; Cornejo-Gaete, C.; Salinas-Neira, B.; Fernández-Bastías, J.; Tupac-Yupanqui, M.; Vidal-Silva, C. Chile-ED-Resp: A Curated Weekly Hospital Dataset for Respiratory Emergency-Demand Forecasting. *Data* **2026**, *1*, 0. <https://doi.org/>

Received:
Accepted:
Published:

Publisher's Note: MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Copyright: © 2026 by the authors. Submitted to *Data* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Abstract: Respiratory emergencies create recurrent seasonal pressure on hospital services, yet reproducible forecasting research is often hindered by the effort required to retrieve, interpret, harmonize, and document public administrative data. Chile-ED-Resp is a curated secondary dataset and reproducible processing pipeline derived from the official Chilean Ministry of Health/DEIS dataset “Atenciones de urgencias de causas respiratorias por semana epidemiológica”, produced by the Sistema de Atención Diaria de Urgencias (SADU). The release focuses on the closed 2022–2025 period and retains official hospital emergency-unit records, respiratory cause labels, total counts, five age strata, establishment characteristics, geographic descriptors, and health-service identifiers. Two complementary analysis-ready artifacts are produced: a long table that preserves official count observations and a hospital-week forecasting table centered on high respiratory infection (IRA Alta), with selected respiratory causes pivoted as covariates. Only deterministic derivatives are added, including coverage diagnostics, chronological train/test labels, seasonal encodings, lags, and prior rolling means; official count fields are neither imputed nor smoothed. Reproducibility is supported through source retrieval metadata, SHA-256 checksums, a machine-readable data dictionary, cause and hospital-coverage inventories, automated validation, unit tests, and a GitHub Actions workflow that rebuilds the release from the official Parquet resource. The resource is intended to provide a transparent foundation for forecasting, benchmarking, health-informatics teaching, and secondary analyses while preserving provenance and the licensing conditions of the government source.

Dataset: Chile-ED-Resp, version 1.0.0. Creator: Balcázar-Quimi, W.; Angulo, C.; Cornejo-Gaete, C.; Salinas-Neira, B.; Fernández-Bastías, J.; Tupac-Yupanqui, M.; Vidal-Silva, C. Publisher/location: GitHub, <https://github.com/cvidalmsu/chile-ed-resp-benchmark>. A versioned Zenodo DOI will be added to the archived release before journal publication: 10.5281/zenodo.REPLACE-AFTER-RELEASE.

Dataset License: Curated data files: Creative Commons Attribution-NonCommercial 2.0 Generic (CC BY-NC 2.0), preserving the licensing conditions linked by the official source portal; software: MIT License.

Keywords: emergency department; respiratory illness; hospital demand; time series; forecasting; data curation; DEIS; SADU; reproducibility; FAIR data; Chile

1. Summary

Hospital emergency departments must absorb demand that changes with epidemiological conditions, season, age structure, and local service organization. Respiratory illness is particularly relevant because recurrent winter epidemics can concentrate consultations and hospitalizations within a limited number of weeks. Forecasting approaches have consequently been used to anticipate emergency attendance, bed demand, and severe respiratory hospitalizations [1–4]. Recent systematic evidence also shows growing use of machine-learning and deep-learning methods for emergency-demand prediction, while emphasizing persistent limitations in validation, data documentation, and transportability [5]. In Chile, regional analyses have reported marked temporal variation in emergency consultations and substantial respiratory-season effects [6,7].

Within this setting, the Chilean Ministry of Health makes weekly respiratory emergency data publicly available through the Departamento de Estadísticas e Información de Salud (DEIS). The source dataset, produced from the Sistema de Atención Diaria de Urgencias (SADU), reports consultations and hospitalizations by respiratory cause, establishment, statistical week, and age group [8,9]. Although the source is highly valuable, downstream forecasting studies must repeatedly solve the same data-engineering tasks: retrieving a large Parquet file, harmonizing field names and types, isolating hospital emergency units, separating consultations from hospitalization rows, checking temporal coverage, pivoting causes to hospital-week features, constructing lagged predictors without leakage, and documenting provenance.

From a data-publication perspective, the principal contribution of Chile-ED-Resp is therefore not a new predictive algorithm. Instead, the resource provides a transparent and reusable data infrastructure derived from the official source. A closely related data-publication strategy has recently been demonstrated in *Data* by Voinea and Voinea, who transformed heterogeneous national health-administration workbooks into harmonized artifacts with provenance logs, checksums, dictionaries, and a reproducible Python pipeline [10]. Other recent Data Descriptors likewise emphasize that dataset value depends on clear variable documentation, access conditions, and intended reuse rather than on inferential novelty alone [11]. The design also follows the FAIR principles of findability, accessibility, interoperability, and reusability [12].

Accordingly, Chile-ED-Resp releases two complementary representations of the same official 2022–2025 respiratory emergency information. The first is a curated long-format table that preserves all qualifying hospital emergency-unit records and the official count values. The second is a forecasting-ready hospital-week table that reorganizes selected consultation causes into columns and adds only deterministic time-series features. An explicit coverage criterion identifies establishments suitable for benchmarking rather than hard-coding a small set of hospitals. The repository records the exact dimensions, hospital inventory, cause inventory, source checksum, retrieval timestamp, and validation results for each release, allowing later versions to remain auditable even if the upstream public portal revises historical records.

2. Data Description

2.1. Official Source and Scope

The upstream resource is the government dataset *Atenciones de urgencias de causas respiratorias por semana epidemiológica*, published through `datos.gob.cl` by the Chilean Ministry of Health/DEIS [8]. The active Parquet resource has identifier `ae6c9887-106d-4e98-8875-40bf2b836041`. At the time this descriptor was prepared, the portal reported weekly updates and linked the resource to the Creative Commons Attribution-NonCommercial 2.0 Generic license [13]. Because the current epidemiological year is continuously updated, the curated release intentionally uses the closed

81 four-year interval 2022–2025. Restricting the versioned release to completed calendar
 82 years avoids treating a partially reported current week as a genuine demand decline.

83 The source schema contains 25 variables after normalization to snake-case names.
 84 These fields describe establishment, region, municipality, health service, establishment
 85 type, administrative dependency, level of care, emergency-care type, coordinates, com-
 86 plexity level, year, statistical week, cause code/order, cause label, total count, and five
 87 age-specific count fields. The public schema and representative field types have inde-
 88 pendently been demonstrated using the same DEIS resource in an open data-analysis
 89 workshop, supporting the reproducibility of the field mapping used here [9].

Table 1: Variable families retained from the official DEIS/SADU source. The complete machine-readable dictionary is distributed as `metadata/data_dictionary.csv`.

Family	Main fields	Interpretation
Establishment	<code>establecimiento_codigo</code> , <code>establecimiento_glosa</code> , <code>tipo_establecimiento</code> , <code>nivel_atencion</code> , <code>nivel_complejidad</code>	Official identifiers and service characteristics.
Geography	<code>region_*</code> , <code>comuna_*</code> , <code>latitud</code> , <code>longitud</code>	Administrative and geographic descriptors.
Health service	<code>servicio_salud_*</code> , <code>dependencia_administrativa</code> , <code>tipo_urgencia</code>	Network and emergency-unit descriptors.
Time	<code>anio</code> , <code>semana_estadistica</code>	Official year and statistical/epidemiological week.
Respiratory event	<code>orden_causa</code> , <code>causa</code> , <code>num_total</code>	Cause label and total consultations or hospitalizations in the stratum.
Age strata	<code>num_menor1anio</code> , <code>num1a4anios</code> , <code>num5a14anios</code> , <code>num15a64anios</code> , <code>num65o_mas</code>	Official age-specific counts.

90 2.2. Released Artifacts

91 The repository generates the artifacts summarized in Table 2. Exact row counts,
 92 numbers of establishments, numbers of causes, and file hashes are written at build
 93 time to `metadata/release_metadata.json`, `metadata/hospital_coverage.csv`,
 94 `metadata/causes.csv`, and `metadata/SHA256SUMS.txt`. Storing these quantities
 95 with the release is deliberate: the source is maintained by an external public agency
 96 and may undergo retrospective corrections, whereas a versioned research dataset must
 97 remain traceable to the precise snapshot used to build it.

Table 2: Principal files created by the reproducible release pipeline.

File	Format	Role
<code>chile_ed_resp_hospital_long</code>	Parquet; CSV.gz	Curated long-format hospital emergency-unit observations with official count fields plus provenance and quality-control variables.
<code>chile_ed_resp_ira_alta_forecasting</code>	Parquet; CSV	One row per hospital-week for selected respiratory consultation causes and deterministic forecasting features.
<code>source_manifest.json</code>	JSON	Retrieval timestamp, upstream identifiers, source URL, byte size, SHA-256 checksum, and license metadata.
<code>release_metadata.json</code>	JSON	Version, period, row/column counts, hospital/cause counts, curation rules, and transformation statements.
<code>hospital_coverage.csv</code>	CSV	IRA Alta temporal coverage by establishment and benchmark recommendation flag.
<code>causes.csv</code>	CSV	Inventory of official cause labels, measure type, row count, and reported totals.
<code>validation_checks.csv</code>	CSV	Machine-readable results of structural and quality-control checks.

2.3. Curated Long Table

The long table restricts the official source to the years 2022–2025, establishments classified as Hospital, and records whose emergency type contains the hospital-emergency designation UEH. No respiratory cause is removed at this stage. Consultation and hospitalization rows are distinguished from the official cause label and stored in the derived field `measure_type`; this separation prevents hospitalization counts from being inadvertently interpreted as consultation-demand predictors.

Beyond normalization of column names and coordinate types, official count fields are retained unchanged. Derived fields include a deterministic row identifier, a human-readable year-week label, the fixed source-resource identifier, the sum of the five official age strata, the difference between that age sum and `num_total`, and a quality flag. Any discrepancy between official totals and age-specific fields is reported rather than silently corrected. Likewise, missing source values remain missing; the release does not fill gaps through mean/median substitution, interpolation, or model-based imputation.

2.4. Forecasting-Ready Hospital-Week Table

For reuse in time-series forecasting, consultation rows are mapped to canonical variables using transparent text rules on official cause labels. The current mapping recognizes high respiratory infection (IRA Alta), influenza, pneumonia, acute bronchitis/bronchiolitis, other respiratory causes, identified COVID-19, and unidentified COVID-19 when those labels are present in the source. The mapping does not alter values: where multiple official rows map to the same hospital-week feature, their reported `num_total` values are summed; where a feature is absent, the resulting cell remains missing rather than being replaced with zero.

Hospital selection for benchmarking is coverage-based. For each establishment, the pipeline counts weeks in which IRA Alta is observed and divides this value by the number of distinct IRA Alta year-week pairs available nationally within the release window. Establishments with coverage of at least 95% receive

recommended_for_benchmark = True. Researchers can therefore reproduce the recommended cohort or choose a different threshold without changing the curated long table.

Temporal features are constructed after sorting observations by hospital, year, and week. The release provides one-, two-, and four-week lags of IRA Alta, together with a previous-year same-week seasonal lag, trailing four- and 12-week means computed strictly from prior observations, and sine/cosine encodings of statistical week. A default chronological split labels 2022–2024 as training and 2025 as testing. Because lagged and rolling fields use only preceding values, the prepared table avoids look-ahead leakage by construction [14].

3. Methods

3.1. Reproducible Processing Workflow

Figure 1 summarizes the curation workflow. The process begins from the stable government resource identifier rather than from a manually downloaded local copy. The downloader uses repeated HTTP attempts and a command-line fallback, verifies that the retrieved object has a valid Parquet signature, computes a SHA-256 digest, and writes a source manifest before transformation. The raw government file is intentionally excluded from Git version control; the release instead stores its provenance meta-data and redistributes only the curated 2022–2025 artifacts under the source-compatible license.

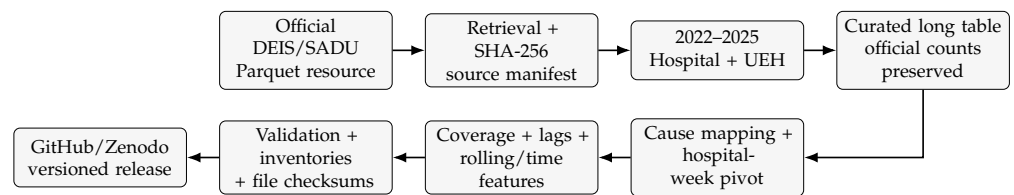


Figure 1. Reproducible Chile-ED-Resp curation workflow. All demand counts originate in the official DEIS/SADU resource; the pipeline adds deterministic organizational, provenance, quality-control, and forecasting fields.

3.2. Schema Harmonization and Type Treatment

Column names are normalized to lower-case snake case. Because public-data schemas can appear with either CamelCase or previously cleaned headings, the build script matches fields through an underscore-insensitive canonical key before checking for the complete expected 25-field schema. If an expected source field is absent, the build stops rather than guessing a replacement. Numeric count, year, week, and cause-order columns are explicitly parsed; coordinates are converted from decimal-comma text to numeric decimal-point representation. No other source values are standardized in a way that changes their substantive meaning.

The curation script also keeps an explicit cause inventory. The inventory records every official cause label present in the selected hospital/UEH period, its consultation/hospitalization classification, number of source rows, and aggregate reported total. Maintaining the original label alongside canonical forecasting features allows users to audit the text mapping and to rebuild alternative targets.

3.3. Hospital Coverage and Selection

Rather than selecting hospitals on the basis of peak demand, geographic convenience, or prior model performance, the release applies a transparent completeness criterion. Let W denote the set of distinct year-week combinations in which IRA Alta is observed anywhere in the hospital subset, and let W_h denote the subset observed for hospital h . Coverage is

$$C_h = \frac{|W_h|}{|W|}. \quad (1)$$

The default benchmark recommendation is $C_h \geq 0.95$. The threshold is not presented as a clinical quality criterion; it is a data-completeness rule intended to reduce avoidable missingness in forecasting comparisons. The full hospital subset remains available so that users may adopt other inclusion criteria.

3.4. Feature Engineering Without Leakage

For a hospital h and ordered weekly target series $y_{h,t}$, lag features are defined as $y_{h,t-k}$ for $k \in \{1, 2, 4, 52\}$. Prior rolling means use

$$\bar{y}_{h,t}^{(m)} = \frac{1}{m} \sum_{j=1}^m y_{h,t-j}, \quad m \in \{4, 12\}, \quad (2)$$

whenever all required prior values exist. The current observation $y_{h,t}$ is excluded from each rolling feature. Seasonal position is encoded as

$$s_t = \sin\left(\frac{2\pi w_t}{52}\right), \quad c_t = \cos\left(\frac{2\pi w_t}{52}\right), \quad (3)$$

where w_t is statistical week. Such deterministic temporal encodings permit use by linear, tree-based, or neural forecasting models without adding external information.

3.5. Automated Build and Versioning

The public repository contains Python scripts for source retrieval, curation, validation, and an optional seasonal-naive baseline. Unit tests cover schema normalization and cause-label mapping. A manual GitHub Actions workflow installs the declared dependencies, executes the tests, downloads the official resource, builds both data artifacts, runs validation, and commits the versioned outputs and metadata. This approach makes the publication package independently rebuildable without requiring authors to distribute the large upstream raw file.

Version-specific provenance is strengthened through cryptographic hashes. The source manifest records retrieval time, file size, and SHA-256 of the official Parquet snapshot; the final checksum file records hashes of the curated Parquet/CSV artifacts and principal metadata files. Similar checksum-based provenance has recently been used in a national health-data Data Descriptor to distinguish an immutable research release from mutable upstream public files [10].

3.6. Technical Validation

Validation is designed to detect structural and source-quality issues without rewriting official observations. Table 3 lists the automated checks. A FAIL status stops the release when the forecasting table cannot be trusted structurally; a WARN status exposes a potentially relevant anomaly while preserving the source values for transparent secondary analysis.

Table 3: Automated validation checks implemented in `scripts/validate_dataset.py`.

Check	Severity	Purpose
Official source counts pre-served	Pass assertion	Confirms that demand counts are read from DEIS/SADU and only derived fields are computed.
Analysis years	Fail if invalid	Ensures the versioned release is restricted to 2022–2025.
Non-negative counts	Warning	Reports unexpected negative administrative counts without silently changing them.
Age-total reconciliation	Warning	Compares <code>num_total</code> with the five age-stratum fields and reports mismatches.
Unavailable age reconciliation	Warning	Reports rows for which the age check cannot be completed because source strata are missing.
Duplicate source strata	Warning	Flags repeated hospital-week-cause keys before wide aggregation.
Unique hospital-week forecast key	Fail if violated	Guarantees one forecasting row per establishment/year/week.
Chronological ordering	Warning	Verifies stable time ordering within the release.
Chronological split	Fail if violated	Ensures training years precede the 2025 test year.
No count imputation	Pass assertion	Documents that missing official counts are not replaced.

In addition to these checks, the pipeline writes the hospital-coverage table and cause inventory so that the cohort and mappings can be inspected independently of the manuscript. An optional seasonal-naive example predicts 2025 IRA Alta from the 52-week lag for recommended hospitals and writes establishment-level MAE, RMSE, and MAPE. Those metrics are intentionally treated as a usage demonstration rather than as a substantive result of the Data Descriptor.

4. User Notes

4.1. Reproducing the Release

Users can reproduce the complete release from a clean checkout with the commands below. Python 3.12 is used in the GitHub Actions workflow, while package versions are constrained through minimum versions in `requirements.txt`.

```
git clone https://github.com/cvidalmsu/chile-ed-resp-benchmark.git
cd chile-ed-resp-benchmark
python -m venv .venv
# activate the environment for the operating system in use
pip install -r requirements.txt
pytest -q
python scripts/make_release.py --start-year 2022 --end-year 2025 --force-download
```

Alternatively, repository maintainers can run *Actions* → *Build DEIS-SADU dataset* → *Run workflow*. The action records the newly retrieved source checksum before rebuilding the release. For a publication snapshot, users should cite the Zenodo DOI associated with the tagged GitHub release rather than a mutable branch URL.

4.2. Recommended Forecasting Use

For comparable model experiments, the recommended starting point is the forecasting artifact:

```
data/chile_ed_resp_ira_alta_forecasting.parquet
```

Users should initially restrict the table to establishments carrying the benchmark recommendation flag. The supplied `split` field provides a simple 2022–2024/2025 holdout.

226 Researchers requiring robust model comparison should additionally use rolling-origin
227 or expanding-window validation inside the training period rather than randomly shuf-
228 fling weekly observations [14]. High-dimensional models should be interpreted cau-
229 tiously because model complexity and sample frequency can materially affect perfor-
230 mance in emergency-demand forecasting [5,15].

231 For epidemiological or descriptive analyses, the long table is preferable because
232 it retains official age strata, cause wording, hospitalizations, and establishment meta-
233 data. Users should avoid summing consultation and hospitalization rows into a sin-
234 gle outcome unless the analytical question explicitly requires such an aggregate. The
235 `measure_type`, `causa`, and `orden_causa` fields are provided to make that distinction
236 inspectable.

237 4.3. Licensing, Attribution, and Provenance

238 The official resource page links the source to CC BY-NC 2.0 [8,13]. The curated
239 data files are therefore distributed under the same license, with explicit attribution to
240 the Chilean Ministry of Health/DEIS and an indication that the authors transformed
241 the source through filtering, type normalization, pivoting, and feature engineering. The
242 MIT license applies only to the software. Users intending commercial reuse should not
243 assume that the code license overrides the non-commercial condition attached to the
244 source-derived data.

245 4.4. Limitations and Reuse Hazards

246 Several limitations should guide interpretation. First, the resource contains ag-
247 gregate administrative counts rather than individual patient records; it cannot support
248 patient-level risk prediction, causal inference, or analyses requiring comorbidity tra-
249 jectories. Second, the meaning and completeness of administrative reporting depend
250 on the upstream SADU system and local establishment practices. The pipeline reports
251 anomalies but cannot independently verify whether every source count reflects com-
252 plete clinical activity. Third, retrospective corrections by the public agency can change
253 the current upstream file; source manifests and versioned releases are therefore essen-
254 tial when reproducing a published analysis.

255 Further limitations arise from temporal and modeling choices. The 2022–2025 win-
256 dow was selected to create a closed contemporary period, not to claim that earlier
257 years are unusable. The 95% IRA Alta coverage threshold is a reproducible benchmark-
258 ing rule rather than a universal hospital-inclusion standard. Derived lags and rolling
259 means remain missing when sufficient history is unavailable, and users should decide
260 how their chosen model handles these rows rather than treating absence as zero. No
261 weather, staffing, bed-occupancy, holiday, or socioeconomic variable is added to the
262 core release because such variables originate from different data systems and would
263 require separate provenance and licensing documentation.

264 Finally, the data should not be interpreted as a direct measure of latent popula-
265 tion disease incidence. Emergency consultations reflect both morbidity and healthcare-
266 seeking behavior, local service organization, referral patterns, and reporting practice.
267 Forecasting models built from the resource are therefore best understood as models of
268 reported emergency demand within the represented public-service context.

269 5. Conclusions

270 In summary, Chile-ED-Resp converts a valuable but operationally oriented
271 Chilean public-health resource into a documented, reproducible, and forecasting-
272 ready research dataset without replacing official count observations. The long artifact
273 preserves the source structure needed for transparent secondary analysis, while the
274 hospital-week artifact supplies deterministic features and a coverage-based cohort for
275 time-series benchmarking. Source manifests, checksums, inventories, validation re-
276 ports, tests, and automated rebuilding support traceability from the public DEIS/SADU

resource to each archived release. By separating data curation from claims about the superiority of a particular forecasting model, the resource provides a stronger foundation for subsequent statistical, machine-learning, and health-informatics studies.

Author Contributions: Conceptualization, W.B.-Q., C.A. and C.V.-S.; methodology, C.A., C.C.-G., B.S.-N., J.F.-B. and C.V.-S.; software, B.S.-N., J.F.-B. and M.T.-Y.; validation, C.C.-G., B.S.-N., J.F.-B. and M.T.-Y.; formal analysis, C.A., C.C.-G., B.S.-N. and J.F.-B.; data curation, B.S.-N., J.F.-B. and C.C.-G.; writing—original draft preparation, W.B.-Q., C.A., B.S.-N. and J.F.-B.; writing—review and editing, all authors; visualization, B.S.-N., J.F.-B. and M.T.-Y.; supervision, C.A. and C.V.-S.; project administration, W.B.-Q. and C.V.-S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable. The dataset is derived from publicly available, aggregate administrative health statistics and contains no patient-level records or direct personal identifiers.

Informed Consent Statement: Not applicable. No individual participants were recruited and no identifiable individual-level data were accessed.

Data Availability Statement: The curated dataset, reproducible processing code, machine-readable metadata, validation outputs, and documentation are available at <https://github.com/cvidalmsu/chile-ed-resp-benchmark>. The version submitted with this Data Descriptor will be archived in Zenodo and the version-specific DOI will be inserted before publication. The upstream records are publicly available from the Chilean Ministry of Health/DEIS dataset *Atenciones de urgencias de causas respiratorias por semana epidemiológica*, resource ID ae6c9887-106d-4e98-8875-40bf2b836041. The repository records the retrieval timestamp and SHA-256 checksum of the source snapshot used for each release.

Acknowledgments: The authors acknowledge the Departamento de Estadísticas e Información de Salud (DEIS), Ministerio de Salud de Chile, for maintaining the public SADU respiratory-emergency dataset used as the source of this curated resource.

Conflicts of Interest: The authors declare no conflicts of interest.

References

- Lucini, F.R.; dos Reis, M.A.; da Silveira, G.J.C.; Fogliatto, F.S.; Anzanello, M.J.; Andrioli, G.G.; Nicolaidis, R.; Beltrame, R.C.F.; Neyeloff, J.L.; Schaan, B.D. Man vs. Machine: Predicting Hospital Bed Demand from an Emergency Department. *PLOS ONE* **2020**, *15*, e0237937. doi:10.1371/journal.pone.0237937.
- Duarte, D.; Walshaw, C.; Ramesh, N. A Comparison of Time-Series Predictions for Healthcare Emergency Department Indicators and the Impact of COVID-19. *Applied Sciences* **2021**, *11*, 3561. doi:10.3390/app11083561.
- Albrecht, S.; Broderick, D.; Dost, K.; Cheung, I.; Nghiem, N.; Wu, M.; Zhu, J.; others. Forecasting Severe Respiratory Disease Hospitalizations Using Machine Learning Algorithms. *BMC Medical Informatics and Decision Making* **2024**, *24*, 293. doi:10.1186/s12911-024-02702-0.
- Ackermann, A.E.F.; Fani, V.; Bandinelli, R.; Sellitto, M.A. Short-Term Prediction in an Emergency Healthcare Unit: Comparison Between ARIMA, ANN, and Logistic Map Models. *Forecasting* **2025**, *7*, 52. doi:10.3390/forecast7030052.
- Blanco, J.; Ferreras, M.; Cosido, O. Predictive Modeling of Hospital Emergency Department Demand Using Artificial Intelligence: A Systematic Review. *International Journal of Medical Informatics* **2025**, *207*, 106215. doi:10.1016/j.ijmedinf.2025.106215.
- Toro Castillo, C.; Parraguez Delgado, B.; Pérez, J. Tasa de Consultas Médicas en el Servicio de Urgencias del Hospital Base de Linares, entre los Años 2019–2023 en Chile. *Revista ANACEM* **2024**, *18*, 96–104. doi:10.70536/RevANACEM/V18N1-311024013.
- Díaz, F.; Carvajal, C. Análisis de la Epidemia Estacional por Virus Respiratorios y Virus Respiratorio Sincial (VRS) del Año 2023 Comparado con Cohorte Prepandemia (2015–2019) en Chile. *Andes Pediátrica* **2025**, *96*, 13–22. doi:10.32641/andespediatr.v96i1.5306.
- Ministerio de Salud de Chile, Departamento de Estadísticas e Información de Salud (DEIS). Atenciones de urgencias de causas respiratorias por semana epidemiológica. Portal de Datos Abiertos del Estado de Chile, 2026. Resource ID ae6c9887-106d-4e98-8875-40bf2b836041; accessed 7 August 2026.
- Ministerio de Salud de Chile, Departamento de Estadísticas e Información de Salud (DEIS). Diccionario de datos: Atenciones de urgencias respiratorias semanal. Portal de Datos Abiertos del Estado de Chile, 2024. Accessed 7 August 2026.
- Voinea, D.V.; Voinea, S.A. A Decade of National Patient Experience Feedback in Romania (2016–2026): A Harmonized, Reproducible Open Dataset. *Data* **2026**, *11*, 185. doi:10.3390/data11080185.
- Berth, H.; Richter, P.; Irmischer, L. A Dataset on Current Stress and Dental Anxiety in a Sample of 129 Oral Surgical Patients. *Data* **2026**, *11*, 183. doi:10.3390/data11080183.

-
12. Wilkinson, M.D.; Dumontier, M.; Aalbersberg, I.J.; Appleton, G.; Axton, M.; Baak, A.; Blomberg, N.; Boiten, J.W.; da Silva Santos, L.B.; Bourne, P.E.; others. The FAIR Guiding Principles for Scientific Data Management and Stewardship. *Scientific Data* **2016**, *3*, 160018. doi:10.1038/sdata.2016.18.
 13. Creative Commons. Attribution-NonCommercial 2.0 Generic (CC BY-NC 2.0), 2004. Accessed 7 August 2026.
 14. Hyndman, R.J.; Athanasopoulos, G. *Forecasting: Principles and Practice*, 3 ed.; OTexts: Melbourne, Australia, 2021.
 15. Tuominen, J.; Lomio, F.; Oksala, N.; Palomäki, A.; Peltonen, J.; Huttunen, H.; others. Forecasting Daily Emergency Department Arrivals Using High-Dimensional Multivariate Data: A Feature Selection Approach. *BMC Medical Informatics and Decision Making* **2022**, *22*, 119. doi:10.1186/s12911-022-01878-7.